CALCULUS OF SEVERAL VARIABLES (MATH 252)

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COURSE OUTLINE

1 Functions of Several Variables
1.1 Introduction
1.2 Geometric Representation of Functions
1.2.1 Basic concepts on regions
1.2.2 Level curves and level surfaces
1.3 Limit and Continuity
1.4 Partial Derivatives
1.4.1 Partial Increments , Total Increment and Differentiability
1.4.2 Chain Rule
1.5 Derivatives of Implicitly Functions
1.6 Directional Derivative and Gradient
1.6.1 Geometrical interpretation - tangent vector and tangent plane
1.6.2 Tangent vector and its normal direction
1.6.3 Tangent plane and its normal direction
1.7 Extreme values

OUTLINE

1.8 Taylor's Formula

1.8.1 Method of Lagrange Multipliers

2 Multiple Integrals

2.1 Review on integral of functions of one variable

2.2 Double Integrals

2.2.1 Calculating Double Integrals

2.2.2 Calculating Double integrals Using Polar Coordinates

2.2.3 Multiple integrals using change of variables

2.2.4 Surfaces and their representations

2.2.5 Area of a surface using double integrals

2.2.6 Mass , Moment of inertia and Centre of Gravity

2.3 Triple Integrals

2.3.1 Calculating Triple Integrals

2.4 Applications

FUNCTIONS OF SEVERAL VARIABLES

Definition 1.1

By a **function of several variables,** we mean a function f defined on a subset D of $\mathbb{R}^n$ for some $n \in N$ with values in $\mathbb{R}$, and we write this fact as $f : D \to \mathbb{R}$

The set D on which F is defined is called the **domain** of the function f , and the set of all values of f , ie , the set $\{f(x_1, \cdots, x_n), (x_1, \cdots, x_n) \in D\}$, is called the **range** of f.

A function $f : D \subseteq \mathbb{R}$ with $D \subseteq \mathbb{R}^n$ is also written as

$$z = \{f(x_1, \cdots, x_n) | (x_1, \cdots, x_n) \in D\}$$

and in that case $x_1, \cdots, x_n$ are called the **independent variables** and z is called the **dependent variable**.

Domain

Definition

A relation f which assigns to each point (x, y) in a subset D of $\mathbb{R}^2$, a unique real number is said to be a scalar function of the two real variables. That is

$$z = f(x, y)$$

where $(x, y) \in D$.

The domain D is called a **region** in $\mathbb{R}^2$

The following example are used to illustrate how we may find the domain of functions in $\mathbb{R}^2$

Example

Find the domain of the following function

1. $z = x^2 + y^2$

   Solution

   The domain of z is the set of all values in the (x, y) plane

2. $z = \sqrt{1 - (x^2 + y^2)}$

   Solution

   $D = \{(x, y) : 1 - (x^2 + y^2) \geq 0\}$. Simplified as

   $D = \{(x, y) : x^2 + y^2 \leq 1\}$

3. $z = \dfrac{1}{\sqrt{1 - (x^2 + y^2)}}$

   Solution

   $D = \{(x, y) : x^2 + y^2 < 1\}$

The domain for example 3 is the same for example 2 except at the boundaries. The graph of functions of two variables are often presented as a Surface in the 3D-space. For function of three variables, the domain is a volume set $V \in \mathbb{R}^3$. The graph is called a hyper-surface.

Exercise

Find the domain of the given functions

1. $f(x, y) = \sqrt{x^2 - 2y}$
2. $f(x, y) = \ln(2x - 3y + 1)$
3. $f(x, y, z) = \dfrac{1}{x^2 + y^2 + 4z}$

Geometric Representation of Functions

Example
$f(x, y) = \sqrt{x^2 + y^2}$ defines a function defined on any subset of $\mathbb{R}^3$ to $\mathbb{R}$.

Geometrically a function of two variables represents a surface S in the *3 - dimensional space*, in such a way that the *projection* of S to the $xy-$ plane is the domain D ,and each line parallel to the $z-$ axis and passing through a point in D cuts the surface at one and only one point.

Thus , given a function $f : D \to \mathbb{R}$, the corresponding surface is the set of all points $(x, y, z) \in \mathbb{R}^3$ such that $z = f(x, y)$ with $(x, y) \in D$, i.e.,

$$S = \{(x, y, z) \in \mathbb{R}^3 : z = f(x, y), (x, y) \in D\}$$

Level curves and level surfaces

Definition 1.7

Let f be a function defined on $D \subseteq \mathbb{R}^2$. Then the subset of D of the form

$$\{(x, y) \in D : f(x, y) = c\}$$

are called **level curves** of f , where $c \in \mathbb{R}$.

If f is a function of three variables , defined on $D \subseteq \mathbb{R}^3$, then the subsets of D of the form

$$\{(x, y, z) \in D : f(x, y, z) = c\}$$

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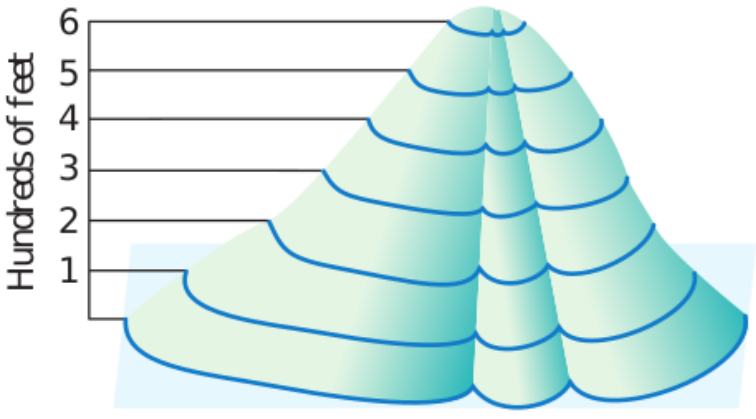
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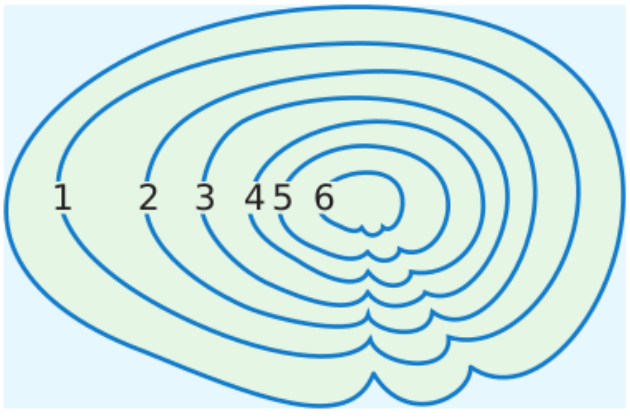
are called **level surfaces** of f, where $c \in \mathbb{R}$.

Note: If f is a function on $D \in \mathbb{R}^2$, then the graph of f is the *surface*,

Level Surface and Level Curve

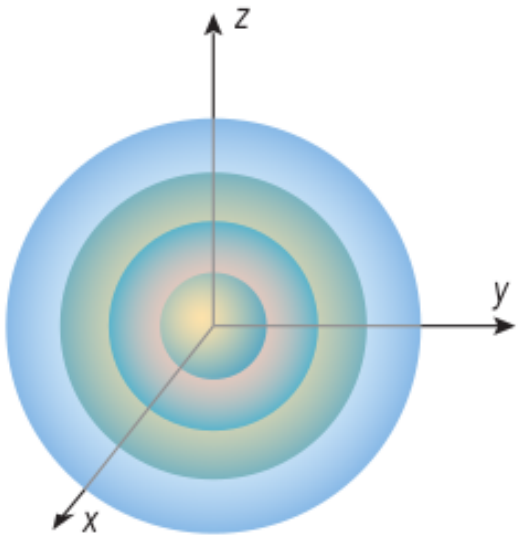


A perspective view of a model hill with two gullies

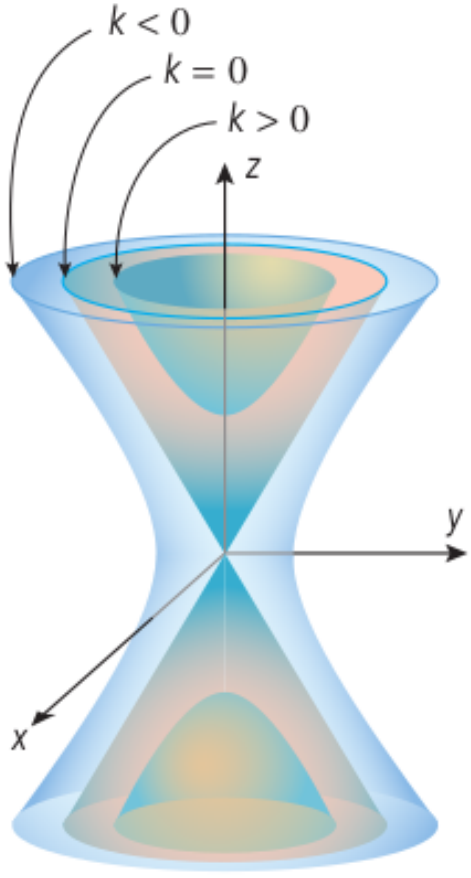


A contour map of the model hill

Level Surface and Level Curve



Level surfaces of
$f(x, y, z) = x^2 + y^2 + z^2$



Level surfaces of
$f(x, y, z) = z^2 - x^2 - y^2$

Level curves and level surfaces

Thus a level curve of f corresponds to the intersection of S_f with the plane $z = c$. The level curve is the projection of the intersection onto the domain of f.

Example Let $f(x, y) = x^2 + y^2$ for $(x, y) \in \mathbb{R}^2$. Then , for $c > 0$, the level curve of f corresponding to c is the circle with centre 0 and radius $\sqrt{c}$. The level curve of f corresponding to $c = 0$ is the point $(0, 0)$, and the level curve of f corresponding to $c < 0$ is the empty set.

Basic concepts on regions (Disk)

Defintion 1.2

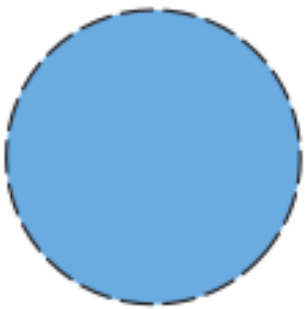
Let $u_0 = (x_0, y_0) \in \mathbb{R}^2$ and $r > 0$.

(1) The set of all points $u = (x, y) \in \mathbb{R}^2$ that satisfy $|u - u_o| < r$ is called the **open disc** with centre $u_0 = (x_0, y_0)$ and radius r.

(2) The set of all points $u = (x, y) \in \mathbb{R}^2$ that satisfy $|u - u_0| \leq r$ is called the **closed disc** with centre $u_0 = (x_0, y_0)$ and radius r.



A closed disk includes all of the points on its bounding circle.



An open disk contains none of the points on its bounding circle.

Boundedness, Interior and Boundary Point

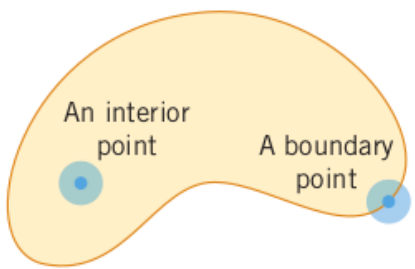
Defintion 1.3

Let D be a region , i.e., a subset of $\mathbb{R}^2$.

(1) D is said to be a **bounded set** if it is contained in a disc (open or closed) for some radius $r > 0$. Sets which are not bounded are called **unbounded** sets

(2) A point (x_0, y_0) is called an **interior point** of D if it is the centre of an open disc contained in D

(3) A point $x_0, y_0)$ is called a **boundary point** of D every open disc containing this point contains some point from D as well as some point not in D



Boundedness, Interior and Boundary Point

(4) The set of all interior points of D is called the **interior** of D , and it is denoted by $int(D)$.

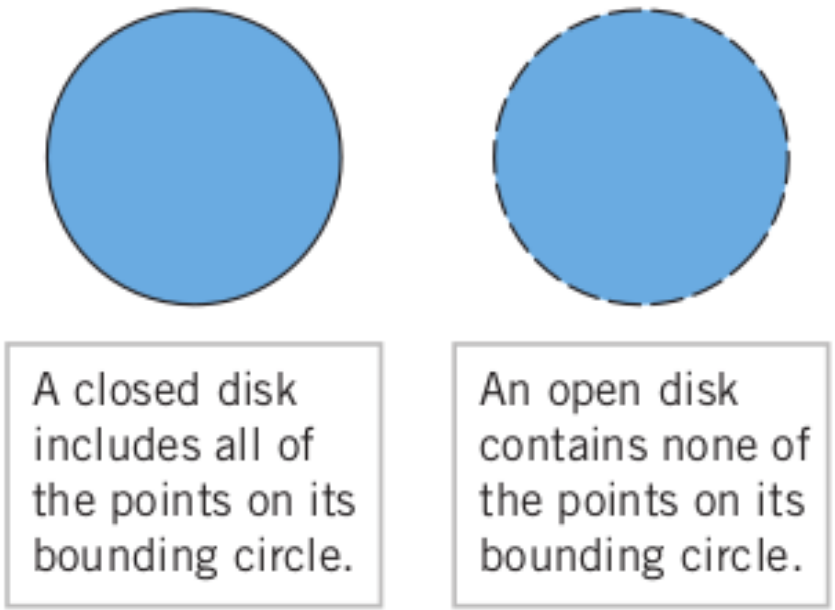
(5) The set of all boundary points of D is called the **boundary** of D, denoted by $bd(D)$.

(6) The set D is called an **open set** it its interior is itself.

(7) The set D is called a **closed set** if it contains all it boundary points.
Thus , for $D \subseteq \mathbb{R}^2$,

- $u_0 \in bd(D) \iff \forall r > o$, $B(u_0, r) \cap D \neq \varnothing$ & $B(u_0, r) \cap \mathbb{R}^2 / D \neq \varnothing$
- D is open $\iff \exists r > 0$, $B(u_0, r) \subseteq D$
- D is closed $\iff bd(D) \subseteq D$.

Example 1.1 An open disc is an open set (*verified*) . If its centre is $u_0 = (x_0, y_0)$ and radius $r > 0$, then its boundary is the circle $(x - x_0)^2 + (y - y_0)^2 = r^2$, i.e., the circle with centre (x_0, y_0) and radius r. Also, the closed disc set contains all its boundary points.



Example 1.2 Let $D = \{(x, y) \in \mathbb{R}^2 : y \leq x^2\}$. This is the region in $\mathbb{R}^2$ below the parabola $y = x^2$, including it. It is a closed unbounded region.

Open Disc and Open Ball

Definition 1.4

Given a particular point $u_0 = (u_1, \cdots, u_k) \in \mathbb{R}^k$ and $r > 0$, the set of all points $x = (x_1, \cdots, x_k) \in \mathbb{R}^n$ that satisfy $|x - u_0| < r$ is called an **open ball** with centre u and radius r, usually denoted by $B(u_0, r)$. Thus ,

$$B(u_0, r) = \{x \in \mathbb{R}^k : |x - u_0| < r\}$$

Open Disc and Open Ball

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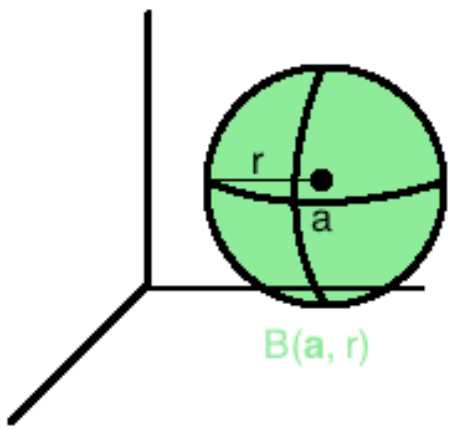


Figure: Open Ball in 3D-Space

Open Disc and Open Ball

Definition 1.5

(a). An open set D is said to be **connected** if any two points in D can be joined by a *line segment* lying in D.

(b). An open connected set is called a **domain**

(c). A set which consists of a domain together with some or all of its boundary points is called a **region**.

Convergence

Definition 1.6

Let (u_n) be a sequence of points in $\mathbb{R}^k$, Then we say that (u_n) **converges** to $u \in \mathbb{R}^k$ if , corresponding to each open ball B centered at u , there is a positive integer N such that $u_n \in B$ for all $n \geq N$.
If (u_n) converges to u, then we write , $u_n \to u$ or

$$\lim_{n \to \infty} u_n = u$$

Limit and Continuity

Definition 1.8

By a **neighbourhood** of a point $u_0 = (x_0, y_0) \in \mathbb{R}^2$ we mean an open disc with centre u_0 and radius δ for some $\delta > 0$, and such that a neighbourhood is also called a δ - **neighbourhood** of $u_0 = (x_0, y_0)$.
A δ- neighbourhood of a point u_0 is usually denoted by $B_\delta(u_0)$.

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Definition 1.9

A neighbourhood with its centre u_0 deleted is called a **deleted neighbourhood** of u_0.
Thus, $\{u \in \mathbb{R}^2 : 0 < |u - u_0| < \delta\}$ is a deleted neighbourhood of u_0.

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Definition 1.10

A point $u_0 \in \mathbb{R}^2$ is called a **limit point** of $D \subseteq \mathbb{R}^2$, if every deleted neighbourhood of u_0 contains at least one point from D.

Limit and Continuity

Example Let $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$. Then $u_0 = (1, 0)$ is a limit point of D , as the sequence (U_n) in D with $u_n = (\frac{n}{n+1}, 0)$ converges to u_0.

Also, the sequence (v_n) in D with $v_n = (\frac{n}{n+1}, \frac{1}{n+1})$ converges to u_0. In fact every point in the *closed disc* $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$ is a limit point of D.

Definition of Limit

Defintion 1.11

Suppose f is defined on a set $D \subseteq \mathbb{R}^2$ and $u_0 = (x_0, y_0) \in \mathbb{R}^2$ is a limit point of D. We say that $f(x, y)$ approaches the **limit** L as $u = (x, y) \in D$ approaches u_o if for every $\epsilon > 0$ there exists a $\delta > 0$ such that

$$|f(u) - L| < \epsilon$$

whenever $u \in D, 0 < |u - u_0| < \delta$.
We write the above fact by

$$\lim_{u \to u_0} f(u) = L, or, \lim_{(x,y) \to (x_0,y_0)} f(x, y) = L$$

as $f(x, y) \to L, or, (x, y) \to (x_0, y_0)$

Definition of Limit

Thus, for a function f defined on a set D, if u_0 is a limit point of D, then
$\lim_{u \to u_0} f(u) = l$ iff for every $\epsilon > 0$, there exists a deleted neighbourhood D_0 of u_0 such that $f(u) \in (l - \epsilon, l + \epsilon)$ for every $u \in D_0 \cap D$.

Theorem 1.0

Let $\lim_{u \to u_0} f(u)$ exists. Then the limit is unique.

Definition of Limit (Example 1)

Let $f(x,y) = \sqrt{1-x^2-y^2}, \quad D = \{(x,y) : x^2+y^2 \leq 1\}$.
Then

$$\lim_{(x,y)\to(0,0)} f(x,y) = 1.$$

Proof:
Let $\epsilon > 0$ be given. If $\epsilon \geq 1$, then $|f(u)-1| = 1-f(u) < \epsilon$ for all $u \in D$. So , let $0 < \epsilon < 1$. Then we have

$$\begin{aligned} |f(u)-1| < \epsilon &\iff 1-\epsilon < f(u) \\ &\iff (1-\epsilon)^2 < 1-|u|^2 \\ &\iff |u|^2 < 1-(1-\epsilon)^2. \end{aligned}$$

Thus , taking $\delta = \sqrt{1-(1-\epsilon)^2} = \sqrt{2\epsilon-\epsilon^2} > 0$ we have $|f(u)-1| < \epsilon$ for $|u| < \delta$.

Definition of Limit (Example 2)

Let

$$f(x,y) = xy\frac{x^2-y^2}{x^2+y^2}, D = \{(x,y) : x^2+y^2 \neq 0\}$$

Then taking $x = r\cos\theta$ and $y = r\sin\theta$ we have $r^2 = x^2+y^2$, and

$$|f(x,y) = |\frac{r^2}{4}\sin 4\theta| \leq \frac{r^2}{4} \to 0, as, r^2 \to 0$$

Thus, for $\epsilon > 0$,

$$|f(x,y)| \leq \frac{|u|^2}{4} < \epsilon$$

whenever $|u| < 2\sqrt{\epsilon}$. Therefore

$$\lim_{(x,y)\to(0,0)} f(x,y) = 0$$

Note that f is not defined at $(0,0)$.

Limits Along Curves

Definition 1.12

If C is a smooth parametric curve in 2-space or 3- space that is represented by the equations

$$x = x(t), y = y(t), or, x = x(t), y = y(t), z = z(t)$$

and if $x_0 = x(t_0), y_0 = y(t_0)$ and $z_0 = z(t_0)$, then the limits

$$\lim_{(x,y)\to(x,y_0)} f(x,y) \text{ and } \lim_{(x,y,z)\to(x_0,y_0,z_0)} f(x,y,z)$$

are defined by

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = \lim_{t\to t_0} f(x(t),y(t)) \tag{1}$$

$$\lim_{(x,y,z)\to(x_0,y_0,z_0)} f(x,y,z) = \lim_{t\to t_0} f(x(t),y(t),z(t)) \tag{2}$$

Limits Along Curves

Note: In these formulas the limit of the function of t must be treated as a one - sided limit if (x_0, y_0) or (x_0, y_0, z_0) is an endpoint of C.

Example: Given that

$$f(x, y) = -\frac{xy}{x^2 + y^2}$$

Find this limits along

(a) the $x-$ axis

(b) the $y-$ axis

(c) the line $y = x$

(d) the line $y = -x$

(e) the parabola $y = x^2$

Solution

(a). The $x-$ axis has parametric equations $x = t$, $y = 0$, with $(0, 0)$ corresponding to $t = 0$, so

$$\lim_{(x,y)\to(0,0)} f(x, y) = \lim_{t\to 0} f(t, 0) = \lim_{t\to 0} \left(-\frac{0}{t^2}\right) = \lim_{t\to 0} 0 = 0$$

Solution

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(b). The $y-$ axis has parametric equations $x = 0$, $y = t$, with $(0, 0)$ corresponding to $t = 0$, so

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(b). The $y-$ axis has parametric equations $x = 0$, $y = t$, with $(0, 0)$ corresponding to $t = 0$, so

$$\lim_{(x,y)\to(0,0)} f(x, y) = \lim_{t\to 0} \left(-\frac{0}{t^2}\right) = \lim_{t\to 0} 0 = 0$$

(c). The line $y = x$ has parametric equations $x = t$, $y = t$, with $(0, 0)$ corresponding to $t = 0$, so

$$\lim_{(x,y)\to(0,0)} f(x, y) = \lim_{t\to 0} f(t, t) = \lim_{t\to 0} \left(-\frac{t^2}{2t^2}\right) = \lim_{t\to 0} \left(-\frac{1}{2}\right) = -\frac{1}{2}$$

Solution Cont

(d). The line $y = -x$ has parametric equations $x = t, y = -t$, with $(0, 0)$ corresponding to $t = 0$, so

$$\lim_{(x,y)\to(0,0)} f(x, y) = \lim_{t\to 0} f(t, -t) = \lim_{t\to 0} \frac{1}{2} = \frac{1}{2}$$

Solution Cont

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(e). The parabola $y = x^2$ has parametric equations $x = t, y = t^2$, with $(0, 0)$ corresponding to $t = 0$, so

$$\lim_{(x,y)\to(0,0)} f(x, y) = \lim_{t\to 0} f(t, t^2) = \lim_{t\to 0} \left(-\frac{t^3}{t^2 + t^4} \right) = \lim_{t\to 0} \left(-\frac{t}{1 + t^2} \right) =$$

This shows that the parametric curve
$x = t, y = t^2, z = -\dfrac{t}{1 + t^2}$
superimposed on the surface.

General Limits Of Functions of Two Variables

Definition 1.13

Let f be a function of two variables , and assume that f is defined at all points of some open disc centered at (x_0, y_0), except possibly at (x_0, y_0). We will write

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L \tag{3}$$

if given any number $\epsilon > 0$), we can find a number $\delta > 0$ such that $f(x,y)$ satisfies

$$|f(x,y) - L| < \epsilon$$

whenever the distance between (x,y) and (x_0, y_o) satisfies

$$0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta$$

Example

$$\lim_{(x,y)\to(1,4)}[5x^3y^2 - 9] = \lim_{(x,y)\to(1,4)}[5x^3y^2] - \lim_{(x,y)\to(1,4)} 9$$

$$= 5\left[\lim_{(x,y)\to(1,4)} x\right]^3 \left[\lim_{(x,y)\to(1,4)} y\right]^2 - 9$$

$$= 5(1)^3(4)^2 - 9 = 71$$

Relationships Between General Limits and Limits along Smooth Curves

Theorem 1.1

(a) If $f(x, y) \to L$ as $(x, y) \to (x_0, y_0)$, then $f(x, y) \to (x_0, y_0)$ alone any smooth curve.

(b) If the limit of $f(x, y)$ fails to exist as $(x, y) \to (x_0, y_0)$ along some smooth curve, or if $f(x, y)$ has different limits as $(x, y) \to (x_0, y_0)$ along two different smooth curves , then the limit of $f(x, y)$ does not exist as $(x, y) \to (x_0, y_0)$.

Continuity

Definition 1.14

A function $f(x, y)$ is said to be **continuous at** (x_0, y_0) if $f(x_0, y_0)$ is defined and if

$$\lim_{(x,y)\to(x_0,y_0)} f(x, y) = f(x_0, y_0)$$

In addition, if f is continuous at every point in an open set D, then we say that f is continuous on D, and if f is continuous at every point in the xy-plane, then we say that f is **continuous everywhere**.

Continuity

Theorem

(a) If $g(x)$ is continuous at x_0 and $h(y)$ is continuous at y_0, then $f(x, y) = g(x)h(y)$ is continuous at (x_0, y_0).

(b) If $h(x, y)$ is continuous at (x_0, y_0) and $g(u)$ is continuous at $u = h(x_0, y_0)$, then the composition $f(x, y) = g(h(x, y))$ is continuous at (x_0, y_0).

(c) If $f(x, y)$ is continuous at (x_0, y_0) , and if $x(t)$ and $y(t)$ are continuous at t_0 with $x(t_0) = x_0$ and $y(t_0) = y_0$, then the composition $f(x(t), y(t))$ is continuous at t_0.

Continuity

Example 1 Evaluate

$$\lim_{(x,y)\to(-1,2)} \frac{xy}{x^2+y^2}$$

Solution

Since $f(x,y) = xy/(x^2+y^2)$ is continuous at $(-1,2)$ (why?), it follows from the definition of continuity for functions of two variables that

$$\lim_{(x,y)\to(-1,2)} \frac{xy}{1-xy} = \frac{(-1)(2)}{(-1)^2+(2)^2} = -\frac{2}{5}$$

Example 2 Since the function $f(x,y) = \dfrac{x^3y^2}{1-xy}$ is a quotient of continuous functions, it is continuous except where $1-xy=0$. Thus, $f(x,y)$ is continuous everywhere except on the hyperbola $xy=1$.

Limits At Discontinuity

Sometimes it is easy to recognize when a limit does exist. For example, it is evident that

$$\lim_{(x,y)\to(0,0)} \frac{1}{x^2+y^2} = +\infty$$

which implies that the values of the function approach $+\infty$ as $(x, y) \to (0, 0)$ along any smooth curve. However, it is not evident whether the limit

$$\lim_{(x,y)\to(0,0)} (x^2+y^2)\ln(x^2+y^2)$$

exists because it is an indeterminate form of type $0.\infty$. Although L'Hospital's rule cannot be applied directly, the following example illustrates a method for finding this limit by converting to polar coordinates.

Example

Find

$$\lim_{(x,y)\to(0,0)} (x^2+y^2)\ln(x^2+y^2)$$

Solution: Let (r,θ) be polar coordinates of the point (x,y) with $r \geq 0$. Then we have $x = r\cos\theta, y = r\sin\theta, r^2 = x^2+y^2$
Moreover, since $r \geq 0$ we have $r = \sqrt{x^2+y^2}$, so that $r \to 0^+$ iff $(x,y) \to (0,0)$.
Thus , we can write the given limit as

$$\lim_{(x,y)\to(0,0)} (x^2+y^2)\ln(x^2+y^2) = \lim_{r\to 0^+} r^2 \ln r^2$$

$$= \lim_{r\to 0^+} \frac{2\ln r}{1/r^2}$$

$$= \lim_{r\to 0^+} \frac{2/r}{-2/r^3} = \lim_{r\to 0^+}(-r^2) = 0$$

Continuity At Boundary Points

For functions of two variables one can extend the notion of continuity of $f(x, y)$ to the boundary of its domain by modifying Definition 1.13 appropriately so that (x, y) is restricted to approach (x_0, y_0) through points lying wholly in the domain of f.

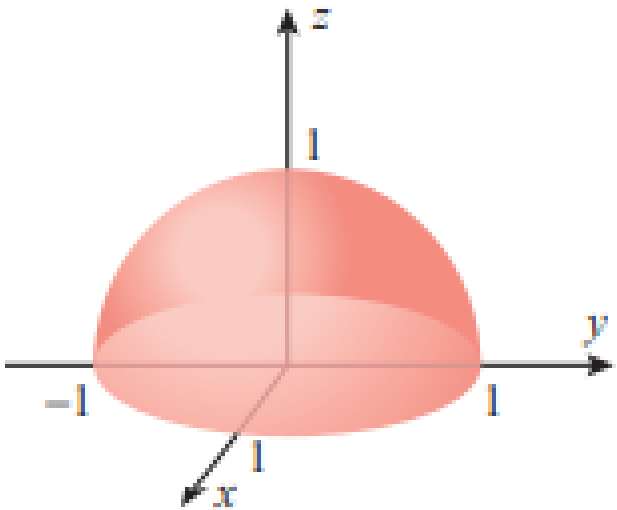
Example: The graph of the function $f(x, y) = \sqrt{1 - x^2 - y^2}$ is the upper hemisphere shown in the Figure below, and the natural domain of f is the closed unit disk $x^2 + y^2 \leq 1$.

Example

The graph of f has no tears or holes, so it passes our "intuitive test" of continuity. In this case the continuity at a point (x_0, y_0) on the boundary reflects the fact that

$$\lim_{(x,y)\to(x_0,y_0)} \sqrt{1 - x^2 - y^2} = \sqrt{1 - {x_0}^2 - {y_0}^2} = 0$$

when (x, y) is restricted to points on the closed unit disk $x^2 + y^2 \leq 1$. It follows that f is continuous on its domain.



EXTENSIONS TO THREE VARIABLES

Definition 1.15

Let f be a function of three variables, and assume that f is defined at all points within a ball centered at (x_0, y_0, z_0), except possibly at (x_0, y_0, z_0). We will writ

$$\lim_{(x,y,z)\to(x_0,y_0,z_0)} f(x, y, z) = L \tag{4}$$

if given any number $\epsilon > 0$, we can find a number $\delta > 0$ such that $f(x, y, z)$ satisfies

$$|f(x, y, z) - L| < \epsilon$$

whenever the distance between (x, y, z) and (x_0, y_0, z_0) satisfies

$$0 < \sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2} < \delta$$

Exercise

1–6 Use limit laws and continuity properties to evaluate the limit.

1. $\lim_{(x,y)\to(1,3)} (4xy^2 - x)$

2. $\lim_{(x,y)\to(0,0)} \dfrac{4x - y}{\sin y - 1}$

3. $\lim_{(x,y)\to(-1,2)} \dfrac{xy^3}{x + y}$

4. $\lim_{(x,y)\to(1,-3)} e^{2x-y^2}$

5. $\lim_{(x,y)\to(0,0)} \ln(1 + x^2y^3)$

6. $\lim_{(x,y)\to(4,-2)} x\sqrt[3]{y^3 + 2x}$

7–8 Show that the limit does not exist by considering the limits as $(x, y) \to (0, 0)$ along the coordinate axes.

7. (a) $\lim_{(x,y)\to(0,0)} \dfrac{3}{x^2 + 2y^2}$ (b) $\lim_{(x,y)\to(0,0)} \dfrac{x + y}{2x^2 + y^2}$

8. (a) $\lim_{(x,y)\to(0,0)} \dfrac{x - y}{x^2 + y^2}$ (b) $\lim_{(x,y)\to(0,0)} \dfrac{\cos xy}{x^2 + y^2}$

9–12 Evaluate the limit using the substitution $z = x^2 + y^2$ and observing that $z \to 0^+$ if and only if $(x, y) \to (0, 0)$.

9. $\lim_{(x,y)\to(0,0)} \dfrac{\sin(x^2 + y^2)}{x^2 + y^2}$

10. $\lim_{(x,y)\to(0,0)} \dfrac{1 - \cos(x^2 + y^2)}{x^2 + y^2}$

11. $\lim_{(x,y)\to(0,0)} e^{-1/(x^2+y^2)}$

12. $\lim_{(x,y)\to(0,0)} \dfrac{e^{-1/\sqrt{x^2+y^2}}}{\sqrt{x^2 + y^2}}$

13–22 Determine whether the limit exists. If so, find its value.

13. $\lim_{(x,y)\to(0,0)} \dfrac{x^4 - y^4}{x^2 + y^2}$

14. $\lim_{(x,y)\to(0,0)} \dfrac{x^4 - 16y^4}{x^2 + 4y^2}$

15. $\lim_{(x,y)\to(0,0)} \dfrac{xy}{3x^2 + 2y^2}$

16. $\lim_{(x,y)\to(0,0)} \dfrac{1 - x^2 - y^2}{x^2 + y^2}$

17. $\lim_{(x,y,z)\to(2,-1,2)} \dfrac{xz^2}{\sqrt{x^2 + y^2 + z^2}}$

18. $\lim_{(x,y,z)\to(2,0,-1)} \ln(2x + y - z)$

19. $\lim_{(x,y,z)\to(0,0,0)} \dfrac{\sin(x^2 + y^2 + z^2)}{\sqrt{x^2 + y^2 + z^2}}$

20. $\lim_{(x,y,z)\to(0,0,0)} \dfrac{\sin\sqrt{x^2 + y^2 + z^2}}{x^2 + y^2 + z^2}$

21. $\lim_{(x,y,z)\to(0,0,0)} \dfrac{e^{\sqrt{x^2+y^2+z^2}}}{\sqrt{x^2 + y^2 + z^2}}$

22. $\lim_{(x,y,z)\to(0,0,0)} \tan^{-1}\left[\dfrac{1}{x^2 + y^2 + z^2}\right]$

23–26 Evaluate the limits by converting to polar coordinates, as in Example 7.

23. $\lim_{(x,y)\to(0,0)} \sqrt{x^2 + y^2}\ln(x^2 + y^2)$

24. $\lim_{(x,y)\to(0,0)} y\ln(x^2 + y^2)$

25. $\lim_{(x,y)\to(0,0)} \dfrac{x^2y^2}{\sqrt{x^2 + y^2}}$

26. $\lim_{(x,y)\to(0,0)} \dfrac{xy}{\sqrt{x^2 + 2y^2}}$

27–28 Evaluate the limits by converting to spherical coordinates (ρ, θ, ϕ) and by observing that $\rho \to 0^+$ if and only if $(x, y, z) \to (0, 0, 0)$.

27. $\lim_{(x,y,z)\to(0,0,0)} \dfrac{xyz}{x^2 + y^2 + z^2}$

END OF LECTURE
THANK YOU